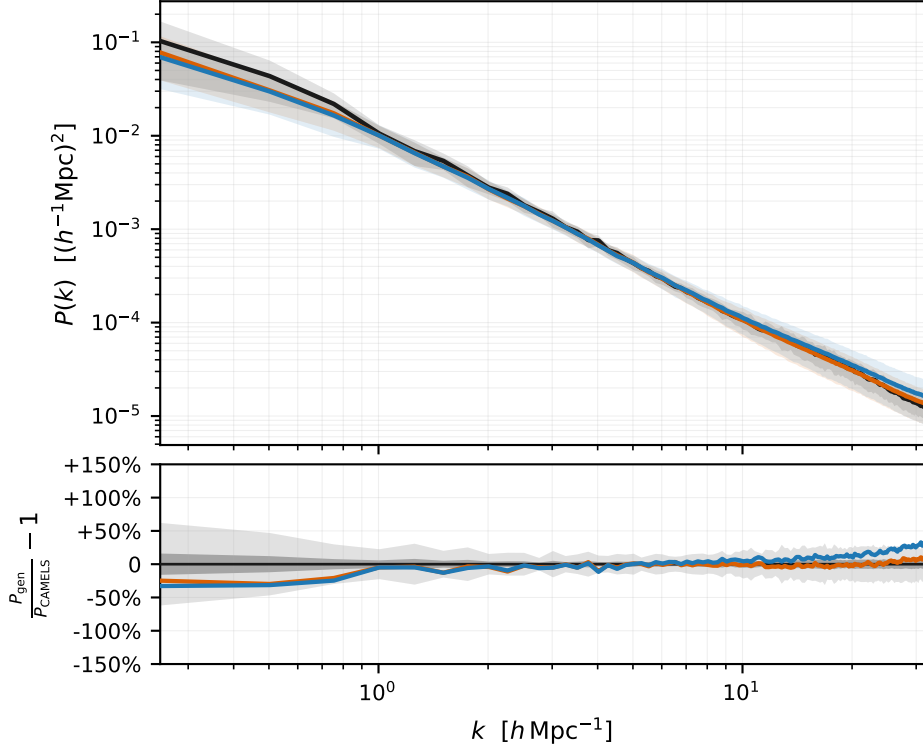


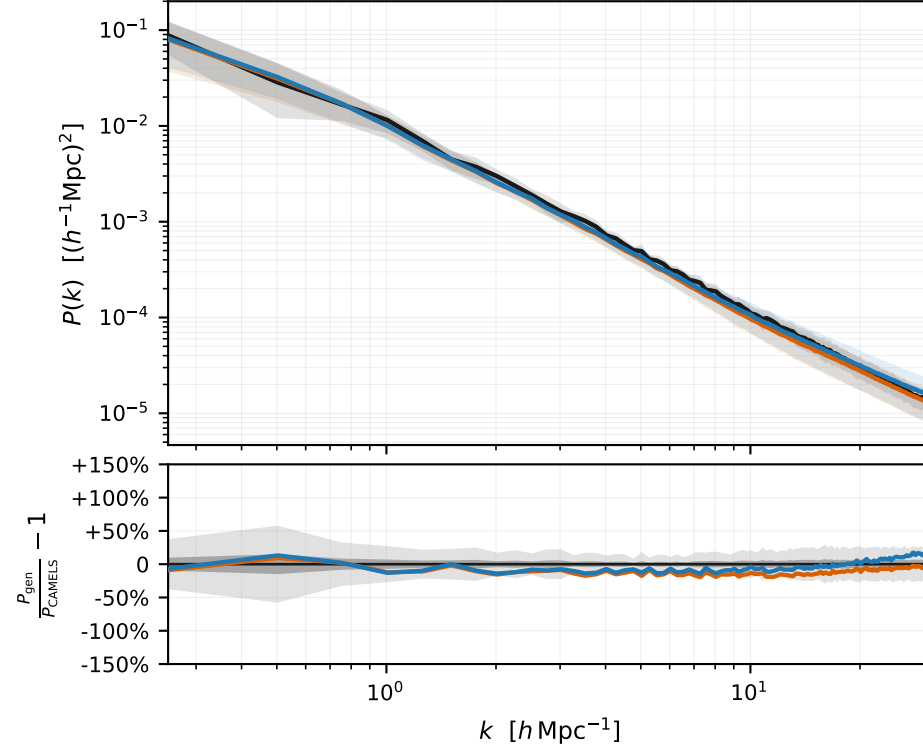
H I column-density power spectrum: CAMELS vs conditional DDPM at six test cosmologies

— CAMELS (15 maps at the same θ) — DDPM-2 (300 samples) — DDPM-6 (300 samples) $\pm 1\sigma$ map-to-map scatter $\pm \sigma/\sqrt{N}$ on the CAMELS mean

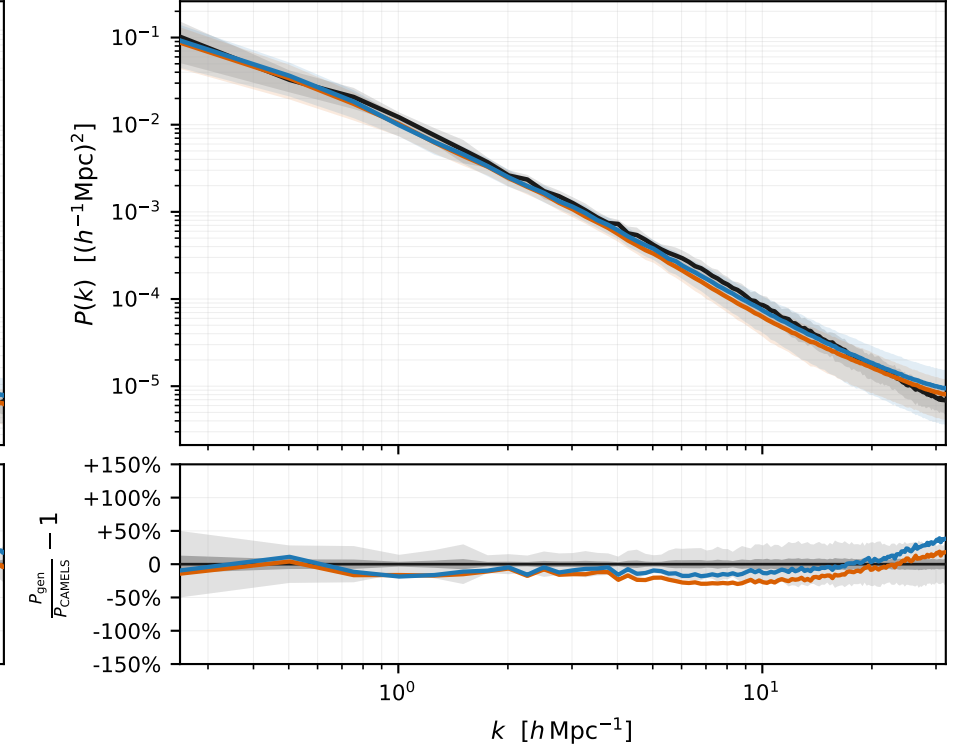
$\Omega_m = 0.329, \sigma_8 = 0.750$
 $A_{\text{SN1}} = 0.34, A_{\text{AGN1}} = 3.30, A_{\text{SN2}} = 1.99, A_{\text{AGN2}} = 1.08$



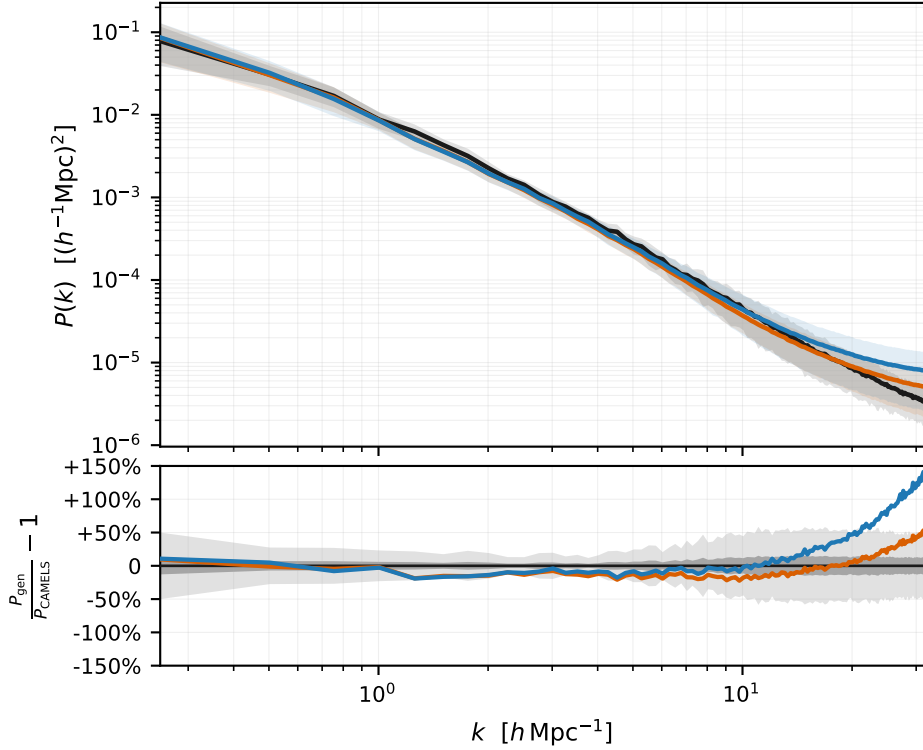
$\Omega_m = 0.282, \sigma_8 = 0.724$
 $A_{\text{SN1}} = 1.59, A_{\text{AGN1}} = 0.27, A_{\text{SN2}} = 1.33, A_{\text{AGN2}} = 0.99$



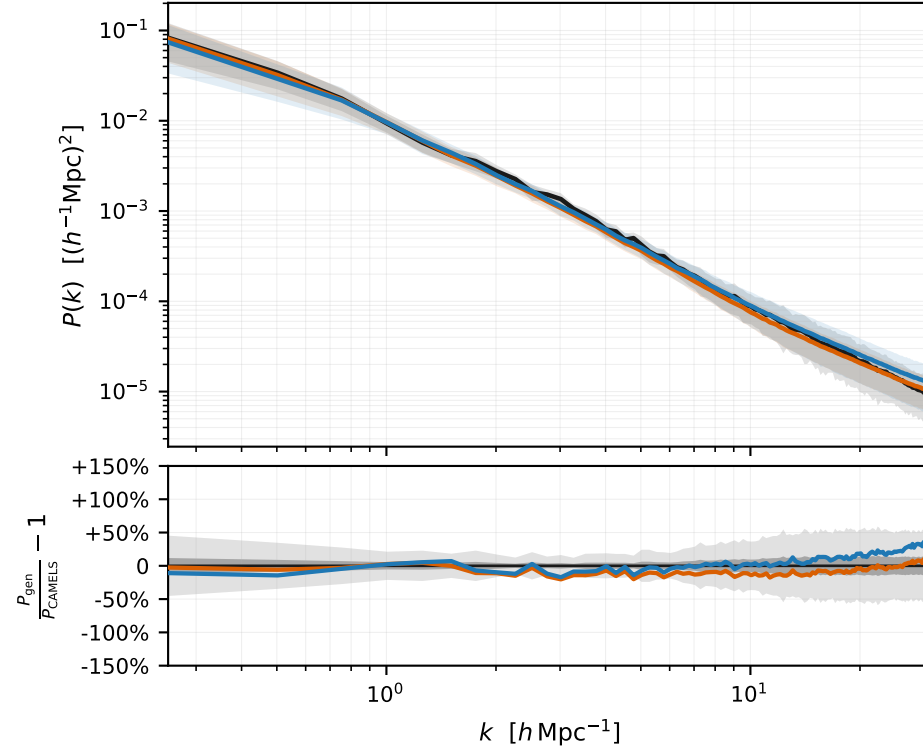
$\Omega_m = 0.161, \sigma_8 = 0.853$
 $A_{\text{SN1}} = 3.40, A_{\text{AGN1}} = 0.36, A_{\text{SN2}} = 0.61, A_{\text{AGN2}} = 1.59$



$\Omega_m = 0.105, \sigma_8 = 0.759$
 $A_{\text{SN1}} = 3.85, A_{\text{AGN1}} = 0.33, A_{\text{SN2}} = 1.93, A_{\text{AGN2}} = 1.81$



$\Omega_m = 0.201, \sigma_8 = 0.747$
 $A_{\text{SN1}} = 0.41, A_{\text{AGN1}} = 1.38, A_{\text{SN2}} = 1.01, A_{\text{AGN2}} = 1.50$



$\Omega_m = 0.247, \sigma_8 = 0.607$
 $A_{\text{SN1}} = 0.70, A_{\text{AGN1}} = 1.34, A_{\text{SN2}} = 1.41, A_{\text{AGN2}} = 0.71$

